

They Prefer Humans! Experimental Measurement of Student Trust in ChatGPT

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ABSTRACT

We investigated student trust in ChatGPT. A multiple choice questionnaire was administered to 171 students. For each question, they chose between one answer from ChatGPT and one from a human expert. Half the answers from ChatGPT and half the answers from the human expert were correct, the other half incorrect. One group saw answers labeled by source. A second group saw unlabeled answers. Participants selected more correct than incorrect answers and showed no preference for incorrect AI answers over correct human answers. We infer that they did not overtrust ChatGPT. However, while the unlabeled group preferred correct AI answers to incorrect human answers, the labeled group did not. We infer that this group undertrusted the technology, probably because of pro-human bias. While we should not underestimate the dangers of overtrust, undertrust may also be a significant issue, depriving students of valuable opportunities.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • Empirical studies in HCI;

KEYWORDS

ChatGPT, Trust, Trust in automation, Undertrust

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1 INTRODUCTION

The rapid evolution of Artificial Intelligence (AI) impacts many aspects of our society, from the economy to education [16]. One recent, important development has been the emergence of so-called

Large Language Models (LLMs). Despite their well-publicized limitations [3], such models' ability to answer questions on a practically unlimited range of topics and to understand, manipulate and generate text, represents a huge advance. At the time of our study, the LLM with the highest number of users, and possibly the best performance is ChatGPT 4.0. ChatGPT 4.0 is accessible to users in Europe for a monthly fee and has received much attention in the educational community [23].

ChatGPT demonstrates LLMs ability to process language and to generate human-like text responses [9, 34]. Unlike conventional chatbots that struggle with context and are limited to specific tasks, it displays a remarkable ability to navigate a broad spectrum of conversational contexts [14]. This versatility stems from its extensive training on a vast dataset from the Internet, a substantial advantage over the narrow, context-bound training of traditional chatbots. ChatGPT also excels in text generation, producing high-quality, diverse, and human-like outputs. In the area of education, these unique capabilities position ChatGPT as a transformative tool, potentially reshaping the way students engage with academic content and opening up new avenues for learning and exploration. For example, [26] show that giving college students the opportunity to use ChatGPT can transform an essay assignment, with learning benefits for students. Similarly, researchers (e.g. [1, 13, 24]) argue that actively interacting in dialogues with ChatGPT can enhance students' involvement with study materials, leading to a more accurate understanding of the subject matter.

While numerous studies have explored ChatGPT's proficiency in generating content, aiding educators, and supporting students in their educational journey (e.g. [7, 24, 27, 32]), few, if any, have investigated how far students trust the technology. [26] show that students would not trust ChatGPT to grade their assignments, believing that the tool still requires human oversight. More generally, [4] suggest that trust is critical to users' adoption of ChatGPT, affecting intended and actual use of the tool. Exploring if and how students trust ChatGPT sheds light on human-AI interaction, with practical implications for responsible deployment and use of LLMs.

Trust in automated technology is a dynamic and context-dependent phenomenon, that evolves through experience and is influenced both by the objective reliability of the technology and by initial levels of trust [12, 15, 17, 18, 31]. In the specific case of AI, trust depends on a range of factors, including the transparency of the system, the way its outputs are presented, as well as the cognitive biases among users [19–21]. In particular, [8] emphasize

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that for AI to obtain the trust of users, it should perform reliably, while exhibiting immediate and comprehensible behavior.

The human-like capabilities, near-instantaneous responses and reliability of ChatGPT may encourage students to trust the system [5]. However, accelerated trust-building could inadvertently lead to overtrust (i.e. reliance on ChatGPT in conditions where it may be unreliable or unsafe). Considering that ChatGPT's proficiency varies across domains [2] and that it can be used to generate persuasive but inaccurate prompts [34], a blind dependence on it could lead to the spread of misinformation. In critical applications such as healthcare, overtrust could even pose risks to human health and safety [4]. The opposite risk, mentioned less often, is that students may undertrust LLMs, choosing not to use them even when they could provide valuable knowledge and services. Properly calibrating students' trust in AI requires a better understanding of their current levels of trust. This exploratory study aims to partially fill this gap in our knowledge. The study has two main objectives:

- **Examine consistent bias:** we investigate whether students demonstrate a bias either towards or against ChatGPT. A consistent bias towards ChatGPT may indicate overtrust in AI-generated information. Conversely, a consistent bias against ChatGPT may indicate undertrust.
- **Evaluate decision-making:** we assess participant decision-making when choosing between answers written by human experts or ChatGPT. In particular, we ask whether students can discern inaccuracies in AI-generated information, a critical factor in preventing the spread of false or potentially harmful content.

2 METHODS

2.1 Experimental design

We built an online multiple-choice questionnaire using the Qualtrics platform. Participants were tasked with answering 20 challenging questions spanning 10 trivia topics: Arts, Literature, Cinematography, Music, Sports, Science, Technology, Geography, History, Economics. The questions, presented in a randomized order, required participants to choose between two answers, where one was written by a Human Expert (HE) and one generated by ChatGPT 4.0 (AI). Half of the human and half of the AI answers were correct, the other half were wrong. The questionnaire presented all possible combinations of correct and incorrect answers in equal proportions (i.e. human correct vs AI correct, human correct vs AI incorrect, human incorrect vs AI correct, human incorrect vs. AI incorrect).

To assess potential biases towards or against ChatGPT, we opted for a between-subjects design. Participants were randomly assigned to two experimental groups, "labeled" and "unlabeled". In the labeled group, the source of the answers (HE or AI) was always presented on screen. In the unlabeled condition, this information was intentionally omitted.

The study was approved by the ethical committee of Leiden University. The data sets, R scripts and prompts used are freely available at: [30].

2.2 Participants

An *a priori* power analysis was performed using G*Power3 [6]. The analysis used a two-tailed test with a medium effect size ($d = .5$)

and an alpha of .05. The analysis indicated that the sample size necessary to reach a power of .80 was 156.

Research has shown that age significantly influences technology usage patterns [28]. To avoid effects of age, we deliberately selected participants within a relatively narrow age range (i.e. 18 to 26). All participants were actively enrolled in a university program.

Of the initial 250 participants, 79 did not meet the eligibility criteria or gave incomplete responses to the questionnaire. The final sample consisted of 171 students (mean age = 20.12, SD = 2.40). Of these, 126 were female, 35 male, 6 non-binary, 1 preferred not to say and 3 did not choose an option. Students at Leiden University received 1 study credit for their participation.

2.3 Creation of survey questions and answers

2.3.1 Survey questions. To generate the survey questions, we prompted ChatGPT as follows: "Generate 10 high quality questions for the topic [topic]. The questions should be on expert level, and hard to answer for someone without extended knowledge". If we deemed a generated question too easy, we asked ChatGPT to generate a more challenging one. Ultimately, following interrater agreement, we selected the two most challenging questions per topic.

Example question for the topic Music: What are the origins and characteristics of the jazz subgenre known as Bebop?

2.3.2 Human Expert answers. Google search was used to find accurate answers from reputable sources, such as dictionaries, scientific papers and well-known websites. Each answer was summarized in 70 to 80 words, the same length as the answers from ChatGPT. This ensured that participants could not identify the source of the answer from its length.

Incorrect answers were based on correct answers to related but different questions. For example, if the question was "What are the origins and characteristics of the jazz sub-genre known as Bebop?", the incorrect answer described the origins and characteristics of Hard Bop Jazz, a related jazz sub-genre. Like the correct answers, incorrect ones were between 70 and 80 words long.

Example answers:

- **Human correct:** Bebop (or "bop") is a type of small-band modern jazz music originating in the early 1940s. Bebop has roots in swing music and involves fast tempos, adventurous improvisation, complex harmonies and chord progressions, and a focus on individual virtuosity. The name "bebop" originates from the sound of nonsense syllables that scat singers improvised in vocal jazz performances.
- **Human incorrect:** Bebop is a style of jazz that emerged in the United States during the late 1940s that is characterized by a strong blues and gospel influence, suspended and dominant chords, minor keys, as well as a strong driving rhythm with a heavy backbeat, setting up a solid groove. Whereas in other styles of jazz, the mold of the Head was generally played in unison by trumpet and sax, in Bebop, this melody was generally harmonised.

2.3.3 AI answers. Correct AI answers were generated using the following prompt: "Can you provide the correct answer to the following question [question]. Please make sure that the length

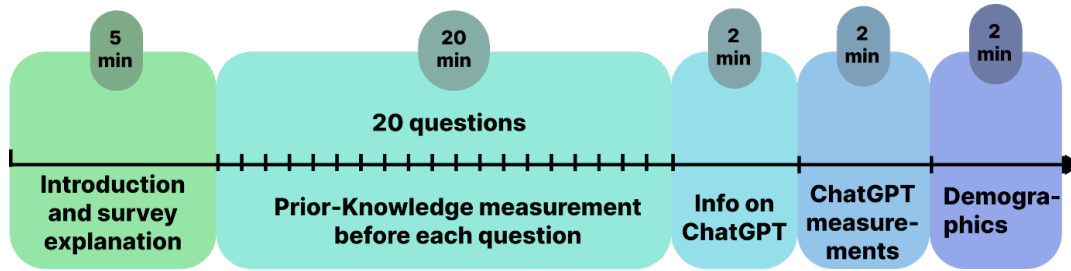


Figure 1: Study procedure.

of the answer is between 70-80 words". Incorrect answers were generated using the prompt: "I want you to generate a false answer to the following question [...]. Note that the correct answer to the question is the following [...]. I want the false answer to be similar to the correct answer, so that it doesn't seem too obvious that this answer is false. Make sure that the false answer is the same length as the text of the answer I provided you (70-80 words)".

Example answers:

- **AI correct:** Bebop, originating in the 1940s, was a jazz sub-genre developed by African American musicians like Dizzy Gillespie, Charlie Parker, and Thelonious Monk as a reaction against swing music's predictable rhythms. Bebop is characterized by complex harmonies, rapid tempos, and improvisation on the melody rather than the chord progression. It's seen as more of an 'artist's music', as its intricate structures made it difficult for casual dancing.
- **AI incorrect:** Bebop, emerging in the 1930s, was a jazz sub-genre introduced by white musicians like Benny Goodman and Glenn Miller as an endorsement of big band swing music's rhythms. Bebop is characterized by simple harmonies, slow tempos, and improvisation based primarily on the chord progression rather than the melody. It's seen more as a 'people's music', as its straightforward structures made it suitable for casual dancing.

2.4 Procedure

Participants who met the eligibility requirements, were presented with the survey instructions. They were informed that the objective was to evaluate their ability to source high-quality, reliable information from the Internet (e.g. Wikipedia, a dictionary, ChatGPT, Google Scholar, etc.). Participants were instructed to select what they considered to be the best answer to each question, even if they were uncertain. Importantly, the instructions stressed that both options might be correct and that participants' should choose what they considered to be the highest quality response.

After providing informed consent, participants were randomly assigned to the "labeled" or the "unlabeled" group. Both groups received the same instructions. .

Prior to answering each question, participants were asked to rate their familiarity with the topic of the question using a Likert scale ranging from 1 (*not familiar at all*) to 5 (*very familiar*). The resulting data will be analyzed in a future manuscript.

After completing all 20 questions, participants were given additional information about ChatGPT (Info on ChatGPT) and asked

to provide information about their previous experience with the tool (ChatGPT measurements). Specifically, they were requested to evaluate how often they had used ChatGPT in the past, their familiarity with the model and its reliability. All evaluations were expressed on 1-5 Likert scales. They were also asked to assess the reliability of human experts, again on a 1-5 Likert scale. Finally, they filled in a form providing personal information, including gender, age, education level, and field of study (Demographics). The entire study lasted approximately 30 minutes (see Figure 1).

3 RESULTS

To account for the multitude of pairwise comparisons in our study (18) and to reduce the risk of Type I errors, we applied the Bonferroni correction method. Consequently, the significance level (alpha) was adjusted to .003.

3.1 Descriptive statistics

Almost half of our participants (47.95%) reported that they had never used ChatGPT and an additional 36.26% had only used it rarely. 60.82% reported no familiarity or low familiarity with the tool (1 - 2 on the Likert scale). Participants were uncertain about the reliability of ChatGPT. 24.56% assessed it to be not at all reliable or only a bit reliable. 42.10% indicated that its reliability was neither good nor bad (Likert: 3). Only 33.33% evaluated it as reliable or completely reliable (Likert 4-5). By contrast, 59.63% of participants rated human experts at this level. This is almost twice the proportion that gave a positive rating to ChatGPT (see Table 1).

3.2 Preference for human or AI answers?

Paired samples t tests were used to investigate whether participants displayed a preference for human or AI answers. Differences between the two groups were explored using independent-samples t-tests.

Across the entire sample, participants chose significantly more human than AI responses, $t(169) = 5.025, p < .001, d = 0.4$. However, further analysis showed that the preference for human responses was restricted to the labeled group $t(90) = 5.88, p < .001, d = 0.6$. No such preference was observed in the unlabeled group, $t(79) = 1.27, p = .208$. These findings indicate a tendency to trust answers from human experts more than answers from AI, but only when the source of answers is known (see Table 2).

Table 1: descriptive statistics of the categorical variables.

	1 Not at all	2	3	4	5 Very
ChatGPT frequency use	82 (47.95%)	62 (36.26%)	11 (6.43%)	11 (6.43%)	5 (2.92%)
ChatGPT familiarity	39 (22.81%)	65 (38.01%)	44 (25.73%)	12 (7.02%)	11 (6.43%)
ChatGPT reliability	2 (1.17%)	40 (23.39%)	72 (42.10%)	55 (32.16%)	2 (1.17%)
Human reliability	2 (1.17%)	7 (4.1%)	60 (35.08%)	91 (53.22%)	11 (6.43%)

Table 2: preference for human or AI answers.

	Sample	Mean Human	SD Human	Mean AI	SD AI	Mean difference	P-value
Human answers vs AI answers (Entire Sample)	171	10.93	2.79	8.99	2.76	1.93	<.001*
Human answers vs AI answers (Labeled)	91	11.51	2.75	8.48	2.68	3.03	<.001*
Human answers vs AI answers (Unlabeled)	80	10.28	2.69	9.58	2.75	0.7	.208

* $p < .003$.**Table 3: preference for correct or incorrect answers.**

	Sample	Mean correct	SD correct	Mean incorrect	SD incorrect	Mean difference	P-value
Correct vs Incorrect (Entire Sample)	171	11.46	2.54	8.47	2.44	2.99	<.001*
Correct vs Incorrect (Labeled)	91	11.66	2.14	8.19	2.36	3.47	<.001*
Correct vs Incorrect (Unlabeled)	80	11.27	2.70	8.71	2.68	2.56	<.001*

* $p < .003$.

3.3 Preference for correct or incorrect answers?

Paired samples t tests were used to investigate whether participants chose more correct than incorrect answers. Across the entire sample, participants chose significantly more correct than incorrect responses, $t(170) = 8.827$, $p < .001$, $d = 0.4$. This preference was present both in the labeled group ($t(90) = 5.034$, $p < .001$, $d = 0.5$), and the unlabeled group ($t(79) = 8.049$, $p < .001$, $d = 0.9$) (see Table 3). Independent samples t tests showed no significant between-groups differences in the numbers of correct (and incorrect) answers chosen (all p 's $> .003$).

3.4 Preference for correct AI or incorrect human answers?

In general, participants in both groups chose more correct than incorrect answers. Surprisingly, however, participants in the labeled group showed no significant difference in preferences between correct AI answers and incorrect human answers ($p > .003$). By contrast, participants in the unlabeled group strongly favored correct AI responses, ($t(79) = 4.074$, $p < .001$, $d = 0.5$).

Independent samples t tests suggest that participants in the unlabeled group chose fewer incorrect human answers compared to participants in the labeled group, but the difference only approached significance, $t(169) = 2.660$, $p = .009$, $d = 0.4$. No difference was

Table 4: preference for correct AI or incorrect human answers

	Sample	Mean AI correct	SD AI correct	Mean Human incorrect	SD Human incorrect	Mean difference	P-value
AI correct vs Human incorrect (Entire Sample)	171	5.57	2.14	5.05	1.98	.53	.04
AI correct vs Human incorrect (Labeled)	91	5.19	2.07	5.42	2.17	-.23	.518
AI correct vs Human incorrect (Unlabeled)	80	6.01	2.14	4.62	1.65	1.39	<.001*

* $p < .003$.

present in terms of the number of correct AI answers chosen by the two groups, $t(169) = -2.559$, $p = .011$.

These findings show that participants could distinguish between correct and incorrect answers. However, in the labeled group this ability was compromised, possibly by pro-human bias (see Table 4). Appendix A.1 presents additional pairwise comparisons which support this conclusion.

4 DISCUSSION

Since the launch of ChatGPT in November 2022, there have been many studies of the intrinsic capabilities and limitations of LLMs. In this paper, by contrast, we focus not on the LLMs themselves, but on how they are used. More specifically, we investigate how far students trust LLMs, and whether the level of trust we observe is “well-calibrated” – a concept borrowed from studies of “trust in automation”. To this end, we developed a multiple-choice questionnaire containing 20 difficult questions on a broad range of non-specialist topics. For each question, we proposed a pair of answers, where one answer came from a human expert and the other from ChatGPT. The questionnaire was administered to 171 students. Half saw answers labeled to show whether they came from AI or from a human source; the other half received unlabeled answers. In both conditions, participants preferred correct over incorrect answers. However, participants in the labeled condition showed a general preference for human answers than answers from ChatGPT. They also chose a greater number of incorrect human answers than incorrect AI answers (see Appendix A.1). Even when the AI answers were correct and the human answers wrong, they showed no clear preference for the right answers. This contrasts with the behavior of participants in the unlabeled condition who consistently chose more correct answers, regardless of source.

Participants in the study were young (age: 18-26) and largely drawn from WEIRD populations [10]. Most reported little previous experience with AI. Future studies with participants from non-WEIRD countries and/or more experienced users may yield different results. A second limitation concerns the questions in the questionnaire. All involved factual information of limited relevance to participants; none touched issues relevant to safety (e.g. health); none required any kind of reasoning; none exploited ChatGPT’s capabilities for essay-writing or for generating computer

code or for analyzing/generating images. We cannot exclude that user responses might differ when faced with more familiar topics and issues. Furthermore, none of the answers were designed to be “persuasive”. In brief, neither questions nor answers were fully representative of real-life LLMs use cases. Finally, it may be argued that we are not measuring students’ trust, but rather their reliance on ChatGPT. While this may be true, previous research has indicated that human behavior can serve as a good proxy for trust in automation (e.g. [22, 29]). In general, trust strongly influences reliance [15, 25, 31], supporting the validity of our conclusions.

In studies of trust in automation, trust is said to be well calibrated when users rely on the automated system in situations where it reliably outperforms humans, but not in situations where it is objectively unreliable [15, 31, 33]. Poor calibration can involve a mix of undertrust and overtrust. Unsurprisingly, participants’ trust in ChatGPT was not perfectly calibrated, i.e. they accepted some AI responses even when they were wrong. Nonetheless, their strong preference for correct over incorrect answers suggests significant resilience to incorrect information, whether from human or AI sources. The hypothesis that users will prefer ChatGPT answers, simply because they are more persuasive than human answers is not supported.

If participants had preferred wrong answers from ChatGPT to correct answers from humans, this would have been a sign of overtrust. However, no such preference was observed. What we did observe was that participants in the labeled condition consistently preferred answers from human experts to answers from ChatGPT – showing no clear preference for the ChatGPT responses even when they were correct and the human answers were wrong. We interpret these findings as undertrust and infer that participants’ ability to distinguish between correct and incorrect answers was confounded by their knowledge of the source of the answers. In short, they suffered from pro-human bias. This notion is supported by participants’ assessment that human experts are more reliable than ChatGPT. In summary, our study found no evidence of overtrust. We did, however, find suggestive evidence of undertrust among participants who were aware of the source of answers.

Many discussions of the use of LLMs in an academic setting focus on the risk that students (and established scholars) will trust the new tools in situations where such trust is inappropriate. The

results of this study point to a different risk - namely that reluctance to use LLMs or outright prohibitions (already introduced by many universities) deprive students (and other scholars) of a potentially valuable tool. This does not mean that overtrust is not a problem - it can be. What our experiment shows is that undertrust (and overtrust in human experts) is an issue in its own right, with its own implications for education and society. We suggest that the most appropriate academic response to the emergence of LLMs is not prohibition, but training for appropriate trust calibration. Previous work on trust in automation suggests that this should involve significant practical experience in the broadest possible set of use cases (e.g. [8, 11, 31]). If users are to successfully navigate LLM's power and limitations, this kind of experiential learning is a necessity. Theoretical instruction (like theory lessons in driving school) is necessary but not sufficient.

Similar arguments can be extended to societal responses to LLM. Here again, overtrust (e.g. in safety-critical situations; in situations where there is risk of discrimination) is a real danger. But undertrust has its own risks - particularly if it leads to inequalities between LLM "haves", whose trust in LLMs is well calibrated to the effective capabilities of the technology, and "have-nots", whose undertrust prevents them from exploiting it, even in situations when it can be useful. Anecdotal evidence suggests that moral panics about the dangers of new technologies (e.g. the printing press, the steam engine, electricity, television, the Internet, social media) are rarely a good guide for public policy. We suggest that in society, as in academia, a more appropriate response is serious theoretical and practical education in the use, strengths, and weaknesses of LLMs. Given the time required to develop appropriate trust calibration, such education should probably start no later than high school.

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A APPENDIX

Table A.1: Complete data for pairwise comparisons of participant answers.

	Sample	Mean AI	SD AI	Mean Human	SD Human	Mean difference	P-value
AI incorrect vs Human incorrect (Entire Sample)	171	3.42	2.08	5.05	1.97	-1.63	<.001*
AI incorrect vs Human incorrect (Labeled)	91	3.30	2.21	5.42	2.17	-2.12	<.001*
AI incorrect vs Human incorrect (Unlabeled)	80	3.56	1.93	4.62	1.65	-1.06	.001*
AI correct vs Human correct (Entire Sample)	171	5.57	2.14	5.88	2.12	-.31	.238
AI correct vs Human correct (Labeled)	91	5.19	2.07	6.09	2.24	-.9	.012
AI correct vs Human correct (Unlabeled)	80	6.01	2.14	5.65	1.97	.36	.34
AI incorrect vs Human correct (Entire Sample)	171	3.42	2.08	5.88	2.12	-2.46	<.001*
AI incorrect vs Human correct (Labeled)	91	3.3	2.21	6.09	2.24	-2.79	<.001*
AI incorrect vs Human correct (Unlabeled)	80	3.56	1.93	5.65	1.97	-2.09	<.001*
AI correct vs Human incorrect (Entire Sample)	171	5.57	2.14	5.05	1.98	.53	.04
AI correct vs Human incorrect (Labeled)	91	5.19	2.07	5.42	2.17	-.23	.518
AI correct vs Human incorrect (Unlabeled)	80	6.01	2.14	4.62	1.65	1.39	<.001*

* $p < .003$.